

Chapter 4

Von Neumann Algebras and The Modular Operator

4.1 Von Neumann Algebras, Concretely and Abstractly

The previous chapter derived Schrödinger dynamics, the Born rule, and Noether's theorem from the intrinsic Kähler geometry of the projective Hilbert space, once a state and a distinguished selfadjoint Hamiltonian were given. The role of the Hamiltonian itself, however, was never questioned. In every construction so far, a time evolution enters the theory as external structure, whether as the one-parameter group of Definition 1.6.23 in the classical setting or as the Hamiltonian flow of Proposition 3.3.2 in the quantum one. This chapter asks whether that structure can instead be recovered from the algebra together with a single state, without postulating a Hamiltonian at all. The Tomita–Takesaki theorem shows that every faithful normal state on a von Neumann algebra determines a canonical one-parameter group of automorphisms, its modular automorphism group. When the state describes thermal equilibrium, this group coincides with ordinary time evolution up to a rescaling by the inverse temperature, an identification that suggests reading the modular flow itself as the origin of physical time whenever no external clock is available. This is the content of the thermal time hypothesis of Connes and Rovelli, and it is the algebraic-geometric standpoint developed throughout this thesis that makes the proposal precise.

Every construction so far in this thesis singled out the C^* -algebra of Definition 1.6.11 as the primitive algebraic object, and derived a Hilbert space representation of an observable algebra only afterward, through the Gelfand-Naimark-Segal construction of Theorem 2.7. The states considered there were arbitrary positive normalized functionals, with no continuity required beyond what automatic continuity of Proposition 2.1.23 already supplies. Tomita–Takesaki theory, together with the notion of a thermal state occupying Section 4.5, requires a strictly stronger setting. The relevant algebra of observables must be weakly closed, so that it coincides with its own bicommutant, and the state that generates the modular flow must be normal, so that expectation values remain continuous under the monotone limits used to define equilibrium. This section introduces von Neumann algebras and normal states as this strengthened setting, and isolates the property under which the Gelfand-Naimark-Segal vector of Theorem 2.7 is simultaneously cyclic for the algebra and separating for it, a distinction the construction of Section 4.3 will use directly. A concrete algebra of operators on a fixed Hilbert space, however, is not the natural datum on the abstract side of the correspondence developed in Chapters 1 and 2, where the observable algebra A of Definition 2.1.22 carries no reference to any particular Hilbert space. The second half of this section therefore isolates the property of A alone, having a predual, that identifies exactly when A can be realized this way, and closes a gap left open by the Gelfand-Naimark-Segal construction itself: nothing in Theorem 2.7 guaranteed that the represented algebra $\pi_\omega(A)$ is weakly closed, hence a genuine von Neumann algebra in the sense about to be introduced.

Definition 4.1.1 (Commutant). *Let H be a Hilbert space, as per Definition 2.2.6, and let $S \subseteq B(H)$ be a subset. The commutant of S is*

$$S' := \{T \in B(H) \mid TA = AT \ \forall A \in S\}.$$

Remark 4.1.2. The commutant S' is a subalgebra of $B(H)$ containing the identity, and every element of S

commutes with every element of S' by definition, so S is contained in its double commutant $S'' := (S')'$.

The characterization of an algebra through its double commutant, rather than through an explicit topological closure, is only useful once the relevant topology on $B(H)$ has itself been fixed. The operator norm topology used throughout this thesis is too fine for this purpose, since it does not detect the monotone limits under which a state is required to behave continuously in Definition 4.1.14 below, so a coarser pair of topologies is introduced first.

Definition 4.1.3 (Weak and strong operator topologies). *Let H be a Hilbert space, as per Definition 2.2.6. The weak operator topology on $B(H)$ is the coarsest topology for which every map $T \mapsto (\phi, T\psi)$, with $\phi, \psi \in H$, is continuous. The strong operator topology is the coarsest topology for which every map $T \mapsto \|T\psi\|$, with $\psi \in H$, is continuous.*

With both topologies available, it becomes possible to ask whether the algebraic condition $M = M''$ used above coincides with topological closure. Von Neumann's original theorem answers this affirmatively for unital $*$ -subalgebras, and this equivalence is what justifies treating the double commutant, an entirely algebraic condition, as the working definition of the topologically closed algebras this chapter needs.

Theorem 4.1 (von Neumann Bicommutant Theorem). *Let $M \subseteq B(H)$ be a unital $*$ -subalgebra, as per Definition 1.6.2. The following are equivalent.*

1. $M = M''$, where M'' denotes the double commutant of Definition 4.1.1.
2. M is closed in the weak operator topology of Definition 4.1.3.
3. M is closed in the strong operator topology of Definition 4.1.3.

This is proved in [TODO-vonNeumannBicommutant].

Definition 4.1.4 (von Neumann algebra). *A von Neumann algebra on H is a unital $*$ -subalgebra $M \subseteq B(H)$, as per Definition 1.6.2, satisfying $M = M''$, as per Definition 4.1.1.*

Remark 4.1.5. By Theorem 4.1, a von Neumann algebra can equally be characterized as a unital $*$ -subalgebra of $B(H)$ closed in either operator topology of Definition 4.1.3. Every von Neumann algebra is in particular a C^* -algebra, as per Definition 1.6.11, under the operator norm, since the norm topology is finer than both operator topologies and closure therefore persists from either of them to the norm topology.

The states relevant to modular theory must interact well with the operator topologies of Definition 4.1.3, not merely with the norm topology already available on any C^* -algebra by Definition 2.1.3. The most direct way to express this compatibility is to exhibit each such state as the restriction of a single linear functional defined on the whole of $B(H)$, built from a fixed trace-class operator rather than from the algebra M itself. This requires first extending the trace of Definition 2.3.15, so far used only for finite-rank density matrices in Proposition 2.3.14, to a linear functional defined on a genuine two-sided ideal of $B(H)$.

Definition 4.1.6 (Trace of a positive operator). *Let H be a Hilbert space, as per Definition 2.2.6, and let $\{e_i\}_{i \in I}$ be an orthonormal basis of H . For $A \in B(H)$ positive, as per Definition 2.2.22, the trace of A is*

$$\text{Tr}(A) := \sum_{i \in I} (e_i, Ae_i) \in [0, \infty].$$

Remark 4.1.7. The value of $\text{Tr}(A)$ in Definition 4.1.6 does not depend on the choice of orthonormal basis $\{e_i\}_{i \in I}$, so that Tr is a well defined map from the positive elements of $B(H)$ to $[0, \infty]$. This invariance is proved in [TODO-traceclass].

Definition 4.1.6 assigns a value in $[0, \infty]$ to every positive operator, but a genuine linear functional requires this value to be finite everywhere, and a genuine ideal of $B(H)$ requires closure not only under sums of positive elements but under multiplication by an arbitrary bounded operator on either side. Both properties hold once the trace is imposed on the modulus of a general, not necessarily positive, operator.

Definition 4.1.8 (Trace-class operators). Let H be a Hilbert space, as per Definition 2.2.6, and let $A \in B(H)$. Write $|A| := (A^*A)^{1/2}$, the positive square root of A^*A as per Remark 2.2.23, positive since $(\psi, A^*A\psi) = \|A\psi\|^2 \geq 0$ for every $\psi \in H$. The operator A is trace-class if $\text{Tr}(|A|) < \infty$, as per Definition 4.1.6. The set of trace-class operators on H is denoted $B_1(H)$.

Remark 4.1.9. The notation $B_1(H)$ parallels the notation $B_0(H)$ of Definition 2.2.15 for the ideal of compact operators. In fact, $B_1(H)$ is itself a two-sided ideal of $B(H)$, contained in $B_0(H)$, and the trace extends from the positive elements of $B_1(H)$ to a linear functional on the whole of $B_1(H)$, satisfying

$$\text{Tr}(AB) = \text{Tr}(BA), \quad A \in B_1(H), B \in B(H).$$

Setting $\|A\|_1 := \text{Tr}(|A|)$ makes $(B_1(H), \|\cdot\|_1)$ a Banach space, as per Definition 1.6.9, and the map

$$B(H) \rightarrow (B_1(H))', \quad T \mapsto (S \mapsto \text{Tr}(ST)),$$

with $(B_1(H))'$ the topological dual of Definition 2.2.13, is an isometric isomorphism. Every fact recalled in this remark is proved in [TODO-traceclass].

The isometric identification of $B(H)$ with the topological dual of $B_1(H)$ singles out a topology on $B(H)$ coarser than the operator norm, and, as the following remark records, also finer than the two topologies of Definition 4.1.3 in general, yet coincident with them on the bounded sets relevant to representing states.

Definition 4.1.10 (Ultraweak topology). Let H be a Hilbert space, as per Definition 2.2.6. The ultraweak topology on $B(H)$ is the coarsest topology for which every evaluation map $\hat{S} : B(H) \rightarrow \mathbb{C}$, $\hat{S}(T) := \text{Tr}(ST)$, with $S \in B_1(H)$ as per Definition 4.1.8, is continuous, generalizing to the Banach space $B_1(H)$ the same weak* construction that Definition 1.6.28 performs on the topological dual of a C^* -algebra.

Remark 4.1.11. The ultraweak topology of Definition 4.1.10 is finer than the weak operator topology of Definition 4.1.3, since the latter only tests continuity against rank-one operators $S = (\cdot, \phi)\psi$, a proper subset of $B_1(H)$. The two topologies coincide on norm-bounded subsets of $B(H)$, so that a von Neumann algebra M , as per Definition 4.1.4, closed in one is automatically closed in the other, consistently with Theorem 4.1 already identifying both with the same purely algebraic condition. This equivalence is proved in [TODO-traceclass].

Restricting the ultraweakly continuous functionals of Definition 4.1.10 from the whole of $B(H)$ to a von Neumann subalgebra M produces a distinguished Banach space attached to M itself, one that supplies the correct, structural, notion of a normal state below.

Definition 4.1.12 (Predual). Let M be a von Neumann algebra on H , as per Definition 4.1.4. The predual of M is the space

$$M_* := \{\omega|_M \mid \omega \in B_1(H)\}$$

of restrictions to M of the trace functionals $S \mapsto \text{Tr}(S \cdot)$, as per Definition 4.1.8, equipped with the quotient norm inherited from $B_1(H)$.

Remark 4.1.13. The predual M_* of Definition 4.1.12 is a Banach space, as per Definition 1.6.9, and the pairing $(\phi, A) \mapsto \phi(A)$, for $\phi \in M_*$ and $A \in M$, identifies M isometrically with the topological dual $(M_*)'$ of M_* , as per Definition 2.2.13, and this identification is unique. Both facts are proved in [TODO-predual].

With the predual in hand, normality of a state can be expressed intrinsically, as membership in M_* , rather than through the monotone limits that only characterize it after the fact.

Definition 4.1.14 (Normal state). Let M be a von Neumann algebra on H , as per Definition 4.1.4. A state ω on M , as per Definition 2.1.3, is normal if $\omega \in M_*$, the predual of M as per Definition 4.1.12.

Remark 4.1.15. Definition 4.1.14 is equivalent to the more classical requirement that

$$\omega\left(\sup_{\alpha} A_{\alpha}\right) = \sup_{\alpha} \omega(A_{\alpha}),$$

for every bounded increasing net (A_{α}) of positive elements of M , with the supremum on the left taken in the strong operator topology of Definition 4.1.3, and equally to continuity of ω for the ultraweak topology of

Definition 4.1.10 restricted to M . This equivalence is proved in [TODO-normalStates]. When $M = B(H)$, it recovers the density-matrix description of Definition 2.3.15: normality of ω is equivalent to the existence of $\rho \in B_1(H)$ positive with $\text{Tr}(\rho) = 1$ such that $\omega(A) = \text{Tr}(\rho A)$ for every $A \in B(H)$, matching Proposition 2.3.14.

Every von Neumann algebra of Definition 4.1.4 carries a predual by construction, since Definition 4.1.12 builds M_* directly from the ambient Hilbert space H . The converse question, whether an abstract C^* -algebra with no reference to any Hilbert space can already be recognized as a von Neumann algebra from the existence of some predual alone, is the content of the abstract characterization due to Sakai, and it is the form in which the correspondence with Chapters 1 and 2 is used from Section 4.2 onward.

Definition 4.1.16 (W^* -algebra). A W^* -algebra is a C^* -algebra A , as per Definition 1.6.11, for which there exists a Banach space A_* , as per Definition 1.6.9, such that A is isometrically isomorphic, as a Banach space, to the topological dual $(A_*)'$ of A_* , as per Definition 2.2.13.

Theorem 4.2 (Sakai). Let A be a W^* -algebra, as per Definition 4.1.16. There exist a Hilbert space H and a $*$ -isomorphism, as per Definition 2.3.1, of A onto a von Neumann algebra M on H , as per Definition 4.1.4. Moreover, the Banach space A_* of Definition 4.1.16 is then unique, and this isomorphism identifies it with the predual M_* of Definition 4.1.12.

This is proved in [TODO-Sakai].

Theorem 4.2 addresses the abstract side of the correspondence, but it leaves open a question specific to the construction already carried out in Chapter 2: nothing in the Gelfand-Naimark-Segal construction of Theorem 2.7 was shown to produce a weakly closed algebra of operators, so $\pi_\omega(A)$ was, until now, only a C^* -subalgebra of $B(H_\omega)$, not yet known to be a von Neumann algebra in the sense of Definition 4.1.4. Normality of the generating state closes exactly this gap.

Proposition 4.1.17 (GNS representations of normal states are weakly closed). Let M be a von Neumann algebra on H , as per Definition 4.1.4, and let ω be a normal state on M , as per Definition 4.1.14. Let $(H_\omega, \pi_\omega, \Psi_\omega)$ be the Gelfand-Naimark-Segal representation of ω , as per Theorem 2.7. Then $\pi_\omega(M)$ is a von Neumann algebra on H_ω , as per Definition 4.1.4.

This is proved in [TODO-GNSnormal].

Remark 4.1.18. Proposition 4.1.17 is what allows Sections 4.2 onward to treat $\pi_\omega(M)$ itself as a concrete von Neumann algebra, with Ψ_ω cyclic for it by part i) of Theorem 2.7, rather than merely as a C^* -subalgebra of $B(H_\omega)$ to which Theorem 4.1 would not yet apply.

The vector Ψ_ω obtained this way is cyclic by construction, and its associated vector state is normal by the following general fact, which holds for an arbitrary vector in a von Neumann algebra without any reference to the Gelfand-Naimark-Segal construction itself.

Remark 4.1.19 (Vector states are normal). Let M be a von Neumann algebra on H , as per Definition 4.1.4, and let $\Psi \in H$ be a unit vector. Setting $\rho := P_\Psi$, the rank-one projector of Definition 2.2.18, the vector state $\omega(A) := (\Psi, A\Psi)$ coincides with the restriction to M of the trace functional $A \mapsto \text{Tr}(\rho A)$, as per Definition 4.1.8, since $\text{Tr}(\rho A) = (\Psi, A\Psi)$ by the same computation as in Proposition 2.3.14. Hence every vector state on M lies in the predual M_* of Definition 4.1.12, and is in particular normal.

Separation for $\pi_\omega(M)$ is a strictly stronger requirement than cyclicity, and it is exactly the combination of both properties in a single vector, on a von Neumann algebra rather than a bare C^* -algebra, that Tomita's construction in Section 4.3 takes as its point of departure.

Lemma 4.1.20 (Cyclic and separating vectors). Let M be a von Neumann algebra on H , as per Definition 4.1.4, and let $\Omega \in H$ be a unit vector, cyclic for M as per Definition 2.2.34. Let ω be the associated vector state, $\omega(A) := (\Omega, A\Omega)$ for $A \in M$. Then Ω is separating for M , meaning that $A\Omega = 0$ with $A \in M$ forces $A = 0$, if and only if ω is faithful, as per Definition 2.3.17.

Proof. Suppose ω is faithful and let $A \in M$ satisfy $A\Omega = 0$. Then $\omega(A^*A) = (\Omega, A^*A\Omega) = \|A\Omega\|^2 = 0$, and faithfulness forces $A^*A = 0$. The C^* -identity of Definition 1.6.11 then yields $\|A\|^2 = \|A^*A\| = 0$, that is, $A = 0$. Conversely, suppose ω is not faithful. There exists a nonzero $B \in M$ with $\omega(B^*B) = 0$, and the same computation gives $\|B\Omega\|^2 = \omega(B^*B) = 0$, so $B\Omega = 0$ with $B \neq 0$. Hence Ω fails to be separating for M . $\square$

Remark 2.3.18 already showed that the Gelfand-Naimark-Segal representation of a faithful state is itself faithful, by tracking the kernel of the representation directly. Lemma 4.1.20 isolates the stronger statement needed here, that the algebra acts on its own cyclic vector without collapsing any nonzero element to zero, since it is this separating property, rather than injectivity of the representation alone, that Tomita's construction in Section 4.3 takes as its starting hypothesis.

4.2 Factors, Projections, and the Trace Problem

Section 4.1 fixed the ambient algebraic setting for the rest of this chapter, a von Neumann algebra M together with a cyclic vector, and the notion of a normal state guaranteeing that expectation values on M interact correctly with the operator topologies of Definition 4.1.3. Before constructing the modular flow itself, it is worth understanding why a construction already available in this thesis, representing a state through a density matrix as in Definition 2.3.15 and Proposition 2.3.14, cannot serve as the general framework this chapter needs. That construction rests on the trace of Definition 4.1.6 being available on the whole of $B(H)$, and on the possibility of writing every normal state as $\omega(A) = \text{Tr}(\rho A)$ for some positive ρ with $\text{Tr}(\rho) = 1$. This section shows that such a trace need not exist on an arbitrary von Neumann algebra at all, and classifies exactly when it does, through the internal structure of the lattice of projections of M . The classification singles out a family of algebras, of type III, on which no nonzero trace of any kind is available, so that density matrices in the sense of Chapter 2 simply do not exist there. It is precisely this obstruction, rather than a technical inconvenience, that motivates replacing the trace with the modular automorphism group constructed in the remainder of the chapter, a substitute that requires only a cyclic and separating vector, as isolated in Lemma 4.1.20, and that reduces to the usual Gibbs description of Section 4.6 whenever a trace happens to be available.

The comparison of projections that drives this classification is carried out through operators that behave isometrically on part of H and vanish on the rest, generalizing the unitary operators already introduced in Definition 2.2.22 to the case where the initial and final spaces need not coincide.

Definition 4.2.1 (Partial isometry). *Let H be a Hilbert space, as per Definition 2.2.6. An operator $U \in B(H)$ is a partial isometry if U^*U is a projector, as per Definition 2.2.17.*

Remark 4.2.2 (Initial and final projections). Let $U \in B(H)$ be a partial isometry, as per Definition 4.2.1, and write $P := U^*U$, the initial projection of U . For every $\psi \in H$,

$$\|U\psi\|^2 = (\psi, U^*U\psi) = (\psi, P\psi) = \|P\psi\|^2,$$

using $P^* = P = P^2$ as per Definition 2.2.17. Hence $\ker U = \ker P$, so that U vanishes on $\ker P = (\text{ran } P)^\perp$ and restricts to an isometry on $\text{ran}(P)$, a closed subspace by Theorem 2.4. Since U vanishes on $\ker P$, it satisfies $UP = U$ as operators on H , so that $Q := UU^*$ satisfies

$$Q^2 = UU^*UU^* = U(U^*U)U^* = UPU^* = UU^* = Q,$$

while $Q^* = Q$ holds trivially. Hence Q is itself a projector, as per Definition 2.2.17, called the final projection of U , with range equal to $\text{ran}(U)$.

Partial isometries supply exactly the notion of comparison needed to declare two projections of a von Neumann algebra equal in size, even when they are not literally the same operator, a comparison that underlies the whole classification to follow.

Definition 4.2.3 (Murray-von Neumann equivalence). *Let M be a von Neumann algebra on H , as per Definition 4.1.4. Two projectors $P, Q \in M$, as per Definition 2.2.17, are Murray-von Neumann equivalent, written $P \sim Q$, if there exists a partial isometry $U \in M$, as per Definition 4.2.1, with $U^*U = P$ and $UU^* = Q$.*

Remark 4.2.4. Definition 4.2.3 defines an equivalence relation on the projectors of M . Reflexivity holds since P itself is a partial isometry with $P^*P = P^2 = P$, so $U = P$ witnesses $P \sim P$. Symmetry holds since U witnessing $P \sim Q$ forces U^* , with $(U^*)^*U^* = UU^* = Q$ and $U^*(U^*)^* = U^*U = P$, to witness $Q \sim P$. For

transitivity, let U_1 witness $P \sim Q$ and U_2 witness $Q \sim R$. Since Q is the final projection of U_1 and the initial projection of U_2 , Remark 4.2.2 yields $QU_1 = U_1$ and $U_2Q = U_2$, so that $U := U_2U_1 \in M$ satisfies

$$U^*U = U_1^*U_2^*U_2U_1 = U_1^*QU_1 = (QU_1)^*U_1 = U_1^*U_1 = P, \quad UU^* = U_2U_1U_1^*U_2^* = U_2QU_2^* = U_2U_2^* = R,$$

so U witnesses $P \sim R$.

Murray-von Neumann equivalence compares projections without reference to any external notion of size, but the classification of factors also needs to compare a projection with a strictly smaller one contained inside it, a containment already implicit in the range inclusions used above.

Remark 4.2.5 (Order on projectors). For projectors $P, Q \in M$, as per Definition 2.2.17, write $P \leq Q$ when $PQ = P$. This holds if and only if $\text{ran}(P) \subseteq \text{ran}(Q)$, and it turns the set of projectors of M into a partially ordered set.

With both a notion of equivalence and a notion of containment available, a projection can be tested against its own equivalents lying strictly inside it, and the outcome of this test is what separates the finite from the infinite case in the trichotomy below.

Definition 4.2.6 (Finite and infinite projections). Let M be a von Neumann algebra on H , as per Definition 4.1.4, and let $P \in M$ be a projector, as per Definition 2.2.17. The projector P is finite if every projector $Q \in M$ satisfying $Q \leq P$, as per Remark 4.2.5, and $Q \sim P$, as per Definition 4.2.3, satisfies $Q = P$. The projector P is infinite if it is not finite.

Having compared projections against each other, the classification turns to the algebra as a whole, through the elements that commute with every one of its own members.

Definition 4.2.7 (Center). Let M be a von Neumann algebra on H , as per Definition 4.1.4. The center of M is $Z(M) := M \cap M'$, with M' the commutant of M , as per Definition 4.1.1.

Remark 4.2.8. The center $Z(M)$ of Definition 4.2.7 is itself abelian. Indeed, for $A, B \in Z(M)$, $A \in M$ and $B \in Z(M) \subseteq M'$ force $AB = BA$, since every element of M' commutes with every element of M by Definition 4.1.1.

Definition 4.2.9 (Factor). A von Neumann algebra M on H , as per Definition 4.1.4, is a factor if $Z(M) = \mathbb{C}1$, with $Z(M)$ the center of M , as per Definition 4.2.7.

The factors are the von Neumann algebras with no nontrivial invariant splitting into independent pieces, and every von Neumann algebra decomposes into a direct integral of factors. The trichotomy below concerns exactly these building blocks. Its statement, however, requires generalizing the finite-dimensional trace already used informally in Definition 2.3.15 to an object that need not be bounded and need not even be finite, so that it can register the difference between a factor on which such a functional exists and one on which it does not.

Definition 4.2.10 (Trace). Let M be a von Neumann algebra on H , as per Definition 4.1.4, and let M_+ denote its positive elements, as per Definition 2.2.22. A trace on M is a map $\tau : M_+ \rightarrow [0, \infty]$ satisfying

$$\tau(A + B) = \tau(A) + \tau(B), \quad \tau(\lambda A) = \lambda \tau(A), \quad \tau(UAU^*) = \tau(A),$$

for all $A, B \in M_+$, all $\lambda \geq 0$, with the convention $0 \cdot \infty := 0$, and all unitary $U \in M$, as per Definition 2.2.22.

Definition 4.2.11 (Faithful trace). Let τ be a trace on M , as per Definition 4.2.10. The trace τ is faithful if $\tau(A) = 0$ with $A \in M_+$ forces $A = 0$.

Definition 4.2.12 (Normal trace). Let τ be a trace on M , as per Definition 4.2.10. The trace τ is normal if

$$\tau\left(\sup_{\alpha} A_{\alpha}\right) = \sup_{\alpha} \tau(A_{\alpha}),$$

for every bounded increasing net (A_{α}) of elements of M_+ , with the supremum on the left taken in the strong operator topology of Definition 4.1.3.

Definition 4.2.13 (Semifinite trace). *Let τ be a trace on M , as per Definition 4.2.10. The trace τ is semifinite if, for every nonzero $A \in M_+$, there exists a nonzero $B \in M_+$ with $B \leq A$, as per Remark 4.2.5 extended to the order induced on all of M_+ , and $\tau(B) < \infty$.*

Remark 4.2.14. Definition 4.2.12 is the same continuity condition as Definition 4.1.14, extended from normalized states to the possibly unbounded functional τ . When $M = B(H)$ and $\tau = \text{Tr}$, as per Definition 4.1.6, restricted to the whole of $B(H)_+$ rather than to $B_1(H)_+$, τ is faithful, normal, and semifinite, and every density matrix ρ of Definition 2.3.15 satisfies $\text{Tr}(\rho) < \infty$, consistently with Definition 4.2.13 applied to $A = 1$.

With finiteness of projections, factoriality, and the trace all in place, the classification of Murray and von Neumann can now be stated. It shows that the existence of a trace on a factor is entirely determined by the combinatorics of its projections, so that the absence of minimal projections already forces a choice between two mutually exclusive possibilities for the whole algebra.

Theorem 4.3 (Murray-von Neumann classification of factors). *Let M be a factor on H , as per Definition 4.2.9. Exactly one of the following holds.*

1. *Type I: M has a minimal nonzero projector; that is, a projector $P \in M$, as per Definition 2.2.17, such that no projector $Q \in M$ satisfies $0 \neq Q < P$ as per Remark 4.2.5. In this case M is $*$ -isomorphic, as per Definition 2.3.1, to $B(K)$ for some Hilbert space K .*
2. *Type II: M has no minimal nonzero projector; but M admits a faithful normal semifinite trace, as per Definitions 4.2.10, 4.2.11, 4.2.12, and 4.2.13.*
3. *Type III: M has no minimal nonzero projector; and M admits no nonzero faithful normal semifinite trace.*

This is proved in [TODO-MurrayVonNeumann].

Remark 4.2.15. Type III factors are not a single, further indistinguishable class. This third case of Theorem 4.3 splits into a family of mutually exclusive subtypes, denoted III_0 , III_λ for $\lambda \in (0, 1)$, and III_1 , detected by an invariant built from the modular operator only once the tools of Section 4.4 are available, and Section 4.9 returns to this finer classification in full. [TODO-fineTypeIII]

The type III case of Theorem 4.3 is where the framework of Chapter 2 breaks down completely, and it is worth spelling out exactly what is lost.

Remark 4.2.16 (Collapse of the density matrix picture). Let M be a factor of type III on H , as per Theorem 4.3. Since M admits no nonzero faithful normal semifinite trace, no normal state ω on M , as per Definition 4.1.14, can be represented as $\omega(A) = \tau(\rho A)$ for a fixed positive $\rho \in M$ with respect to any trace τ on M , in contrast with the description of Definition 2.3.15 and Proposition 2.3.14 available for $B(H)$ itself. Every normal state on M is nevertheless still a vector state for some faithful representation of M , by Remark 4.1.19 together with the Gelfand-Naimark-Segal construction of Theorem 2.7, so a Hilbert space description of the state survives even though a density matrix does not. Algebras of local observables in relativistic quantum field theory are type III on physical grounds, so that this collapse is the generic situation for such algebras rather than a pathological exception. [TODO-QFTtypeIII]

Theorem 4.3 therefore leaves this thesis with a genuine gap. Chapter 2 built the entire probabilistic content of quantum mechanics, states as density matrices and time evolution as conjugation by $e^{-itH/\hbar}$, on the trace of $B(H)$, a type I factor by Theorem 4.3 applied with $M = B(H)$ itself, since every rank-one projector is minimal there. On a type III factor, neither notion is available in that form. The remainder of this chapter constructs, for an arbitrary von Neumann algebra M together with a vector cyclic and separating for it as in Lemma 4.1.20, a canonical one-parameter group of automorphisms of M that requires no trace whatsoever, built instead directly from the vector state itself through Tomita's theorem of Section 4.4. Section 4.6 will verify explicitly that, when M does possess a trace, this group reduces to ordinary Gibbs time evolution, so that the construction to follow genuinely extends, rather than replaces, the familiar type I picture of Chapter 2.